

Data Analysis and Machine Learning Report

1 Introductions

This report presents an analysis of the *Kaggle Sales Data Sample* dataset and the application of machine learning models. The objectives are to identify trends, classify sales, and predict sales values. Visualizations and model results are detailed throughout the report.

2 Data Preparation

Data

The *Kaggle Sales Data Sample* dataset contains transactional records from a fictional sales database. It provides detailed information about orders, customers, products, and sales performance metrics. Key aspects of the dataset are as follows:

Orders:

Each order is uniquely identified by an ORDERNUMBER.

Products

The dataset includes a diverse range of products categorized by PRODUCTLINE. Additional product details include the *PRODUCTCODE* for unique identification, the *MSRP* (Manufacturer's Suggested Retail Price) for pricing information, and the *DEALSIZE*, which indicates the scale of each transaction.

Customers

Customer details such as CUSTOMERNAME, CITY, COUNTRY, and CONTACT information provide insights into the geographic and demographic diversity of the clientele.

Sales Performance

The SALES column records the monetary value of each transaction, which is the primary target variable for prediction tasks.

Related metrics include QUANTITYORDERED (the number of units purchased) and PRICEEACH (unit price).

Time-Based Features

The dataset captures temporal aspects such as ORDERDATE, MONTH_ID, and YEAR_ID, allowing for time-series analysis and trend identification.

Additional Information

Columns such as STATUS describe the order's lifecycle stage (e.g., shipped, cancelled).

Address details (ADDRESSLINE1, ADDRESSLINE2, STATE, POSTALCODE) enrich the dataset for location-based analysis.

Loading and Cleaning

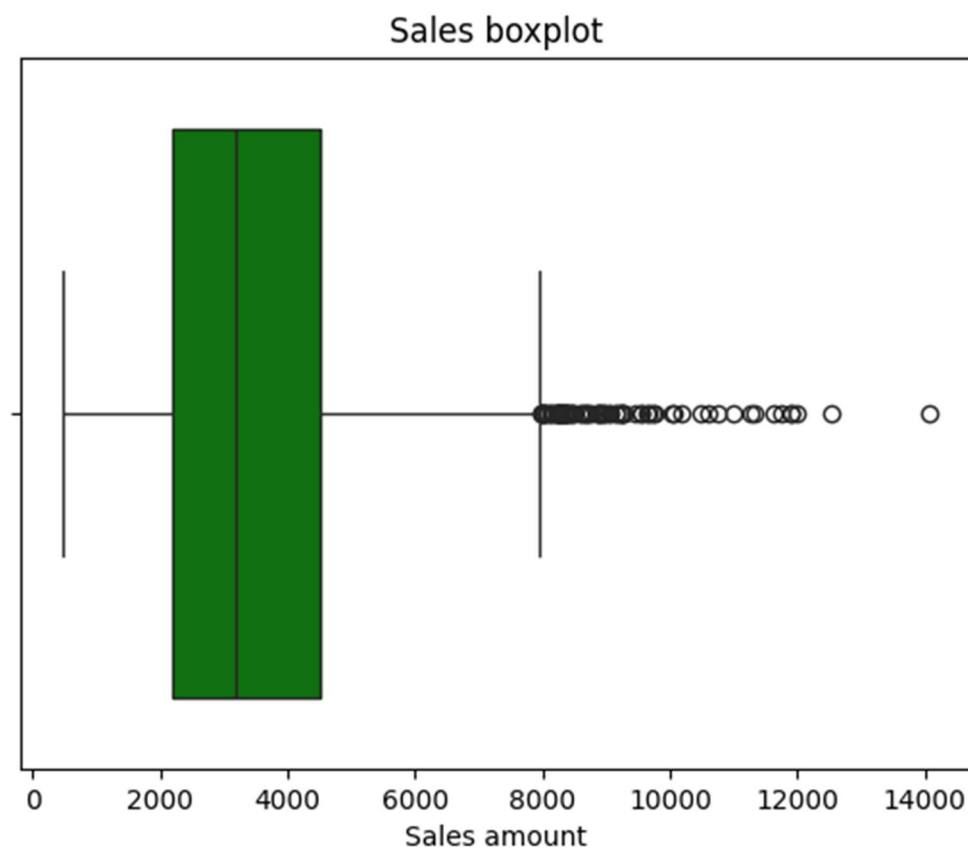
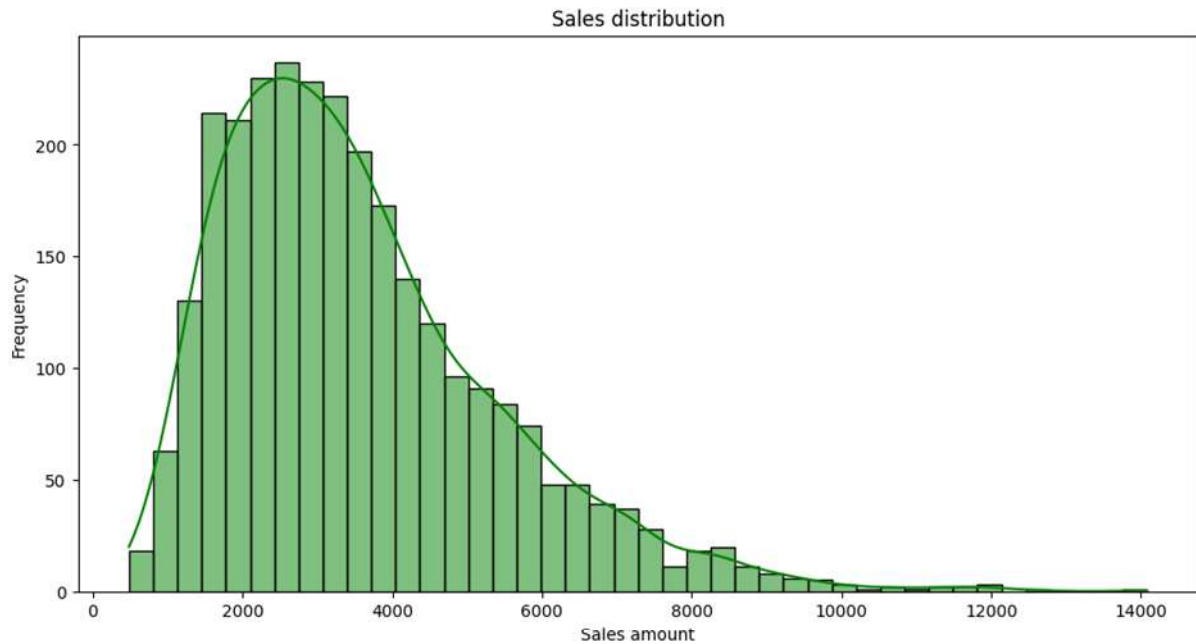
Missing values were replaced with zero, as the missing data primarily consisted of address-related fields that were not relevant to our analysis.

Additionally, several unnecessary columns were removed, such as *ORDERNUMBER*, *PHONE*, and *ADDRESSLINE2*, to streamline the dataset and focus on relevant features. By removing these columns, the dataset was streamlined to focus on key features that drive sales trends and predictions.

Save Dataset

The preprocessed data was saved for further analysis.

3 Visualization



Histograms and boxplots show that most sales are low, with some significant outliers.

Highest Sale

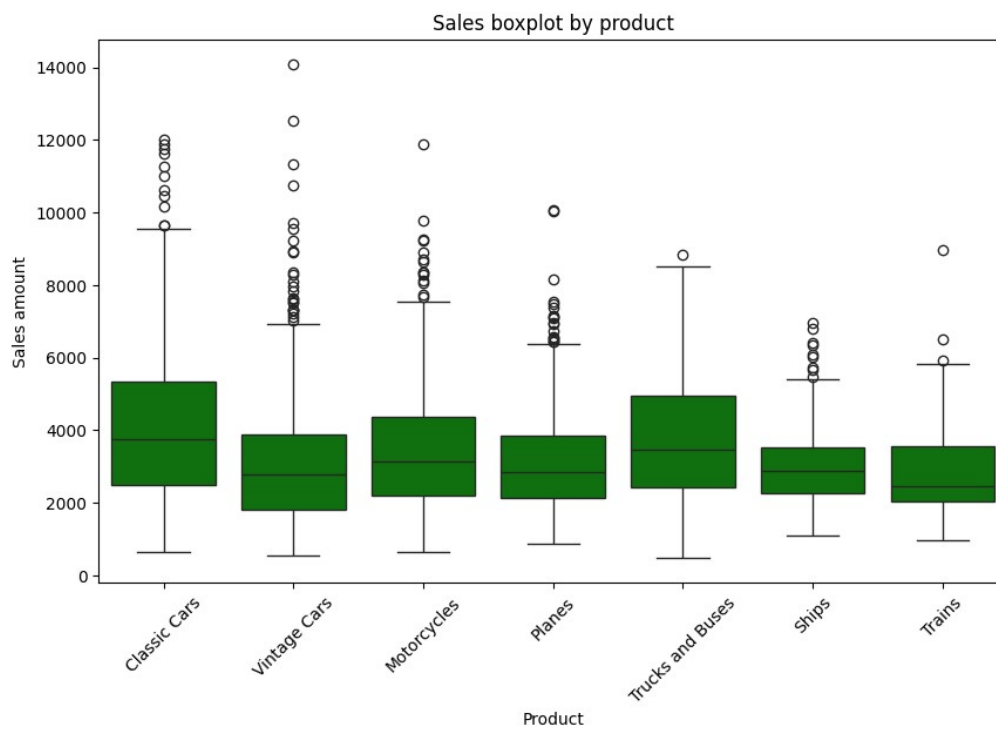
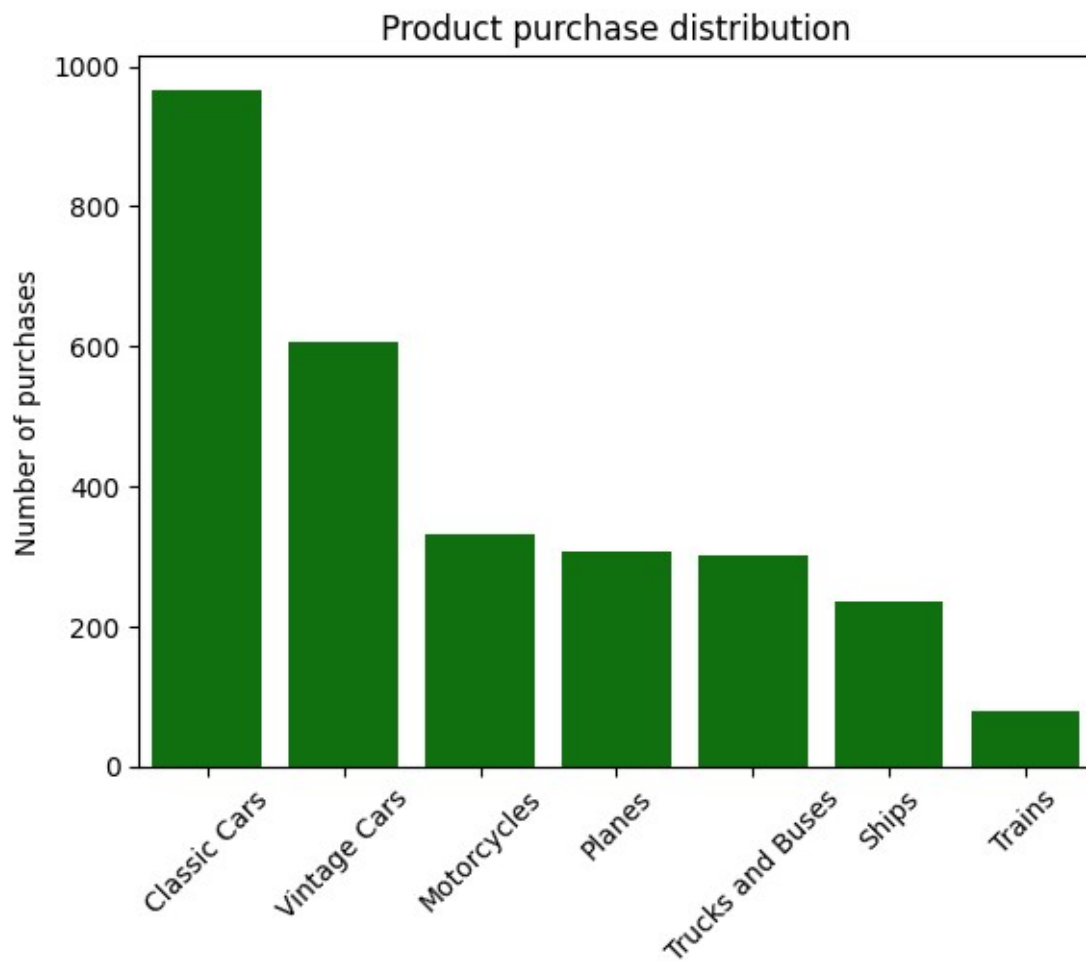
The most expensive sale was highlighted:

SALES	14082.8
ORDERDATE	2005-04-22 00:00:00
PRODUCTLINE	Vintage Cars

CITY
COUNTRY

San Jose
USA

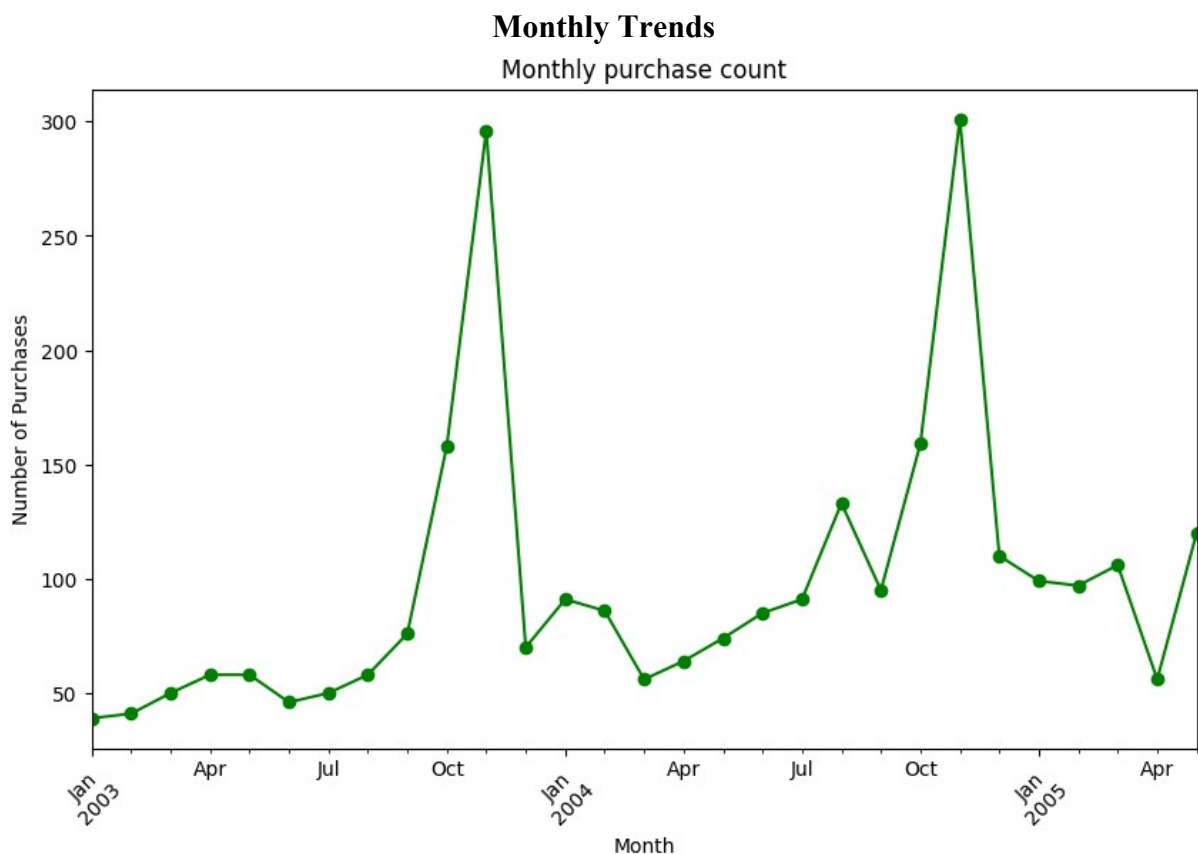
Product Category Distribution



Sales counts by product categories were visualized using a bar chart to analyze purchasing trends across different product lines. The chart revealed that the majority of purchases were concentrated in the *Vintage Cars* and *Classic Cars* categories, indicating their popularity among customers. These categories had the highest number of transactions, suggesting they may be key revenue drivers for the business.

In contrast, the *Trains* and *Ships* categories had the fewest purchases, highlighting their lower demand compared to other product lines. This disparity could reflect a niche market or limited customer interest in these items.

This visualization provides valuable insights into product performance, helping to identify top-selling categories while also spotlighting underperforming areas that may require targeted marketing efforts or reevaluation of inventory strategies.



Temporal analysis of sales counts revealed a distinct seasonal pattern, with a notable peak in November. This surge in sales is likely attributed to the holiday season, particularly the increased consumer activity surrounding Christmas.

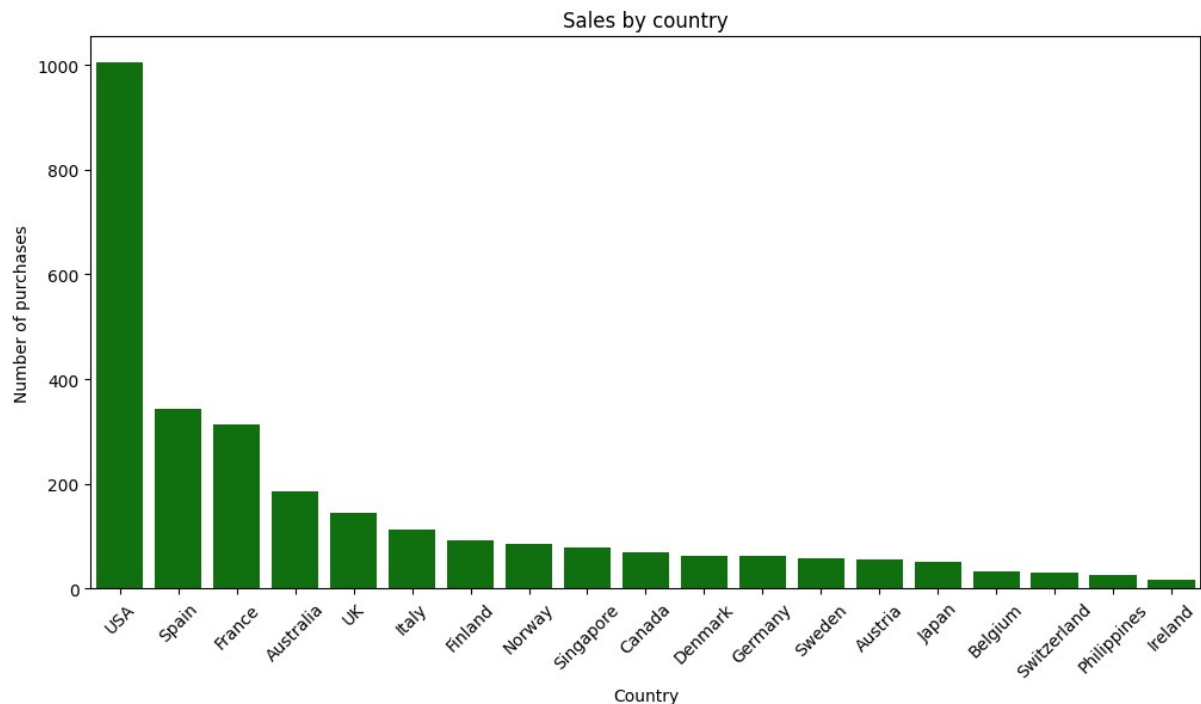
Understanding this seasonal behavior is crucial for optimizing inventory management, marketing strategies, and promotional campaigns to capitalize on the increased demand during this time of year.

Sales by Country

Geographic distribution of sales was analyzed through a bar chart.

As shown in the sales by country visualization, the USA leads in sales, followed by Spain in second place. This suggests that these countries represent the largest markets for the products. The most interesting opportunity lies in further exploring the market dynamics in these regions,

particularly the USA, to better understand customer preferences and tailor marketing strategies accordingly.



4 Classification

In this analysis, the goal was to classify the sales data into distinct categories based on sales performance. To achieve this, the dataset was divided into three categories: *Low*, *Medium*, and *High*, based on the sales values. This classification was performed using a machine learning approach to predict the sales category of each transaction based on various features.

Feature Selection and Preprocessing

Several important features were selected for classification, including *QUANTITYORDERED*, *PRICEEACH*, *MONTH_ID*, *YEAR_ID*, *STATUS*, *PRODUCTLINE*, *CITY*, *COUNTRY*, and *DEALSIZE*. These features were used to build the input variables for the model.

In addition, categorical variables such as *STATUS*, *PRODUCTLINE*, *CITY*, *COUNTRY*, and *DEALSIZE* were encoded using Label Encoding to make them suitable for machine learning algorithms.

Model Selection and Hyperparameter Tuning

For the classification task, a Random Forest Classifier was chosen due to its robustness in handling both categorical and numerical data. Hyperparameter tuning was performed using *RandomizedSearchCV*, where the model's parameters were optimized for the best accuracy. This process involved testing different values for parameters such as *max_depth*, *n_estimators*, *criterion*, and others to find the optimal combination for the classifier.

Model Performance

After training the **Random Forest Classifier**, the best hyperparameters were identified through the **RandomizedSearchCV**, which optimized the model for the best possible performance. The optimal parameters found were:

- **Criterion:** 'entropy'
- **Max Depth:** 10
- **Max Samples:** 0.86
- **Min Samples Split:** 0.011
- **Number of Estimators:** 100

With these settings, the model achieved an accuracy of 88.5%, indicating a strong ability to correctly classify the sales data into Low, Medium, and High categories. This high accuracy demonstrates that the model is effective in predicting sales performance and can be used for making informed decisions about sales strategies and inventory management.

Insights

The classification model is valuable for understanding how sales transactions can be grouped based on their monetary value. By categorizing sales, businesses can identify trends, optimize inventory, and target specific customer segments more effectively. The model's predictions can also inform sales strategies, helping to prioritize high-value transactions or identify areas that need improvement.

5 Regression

In addition to the classification analysis, a regression model was implemented to predict the actual sales values, providing insights into the continuous nature of sales performance. This regression task, which aimed to predict the *SALES* column using various features from the dataset, enables the estimation of sales amounts based on input variables. This approach is the **culmination** of the analysis, as it directly addresses the primary goal of predicting sales figures, making it the most relevant and fitting method for this task. The regression model provides actionable insights into sales performance and can be used for more accurate forecasting and decision-making.

Feature Selection and Preprocessing

For the regression model, the same features used in the classification model were selected, including *QUANTITYORDERED*, *PRICEEACH*, *MONTH_ID*, *YEAR_ID*, *STATUS*, *PRODUCTLINE*, *CITY*, *COUNTRY*, and *DEALSIZE*. These features were treated as predictors for the regression model. Additionally, categorical variables were encoded using Label Encoding to convert them into numerical values suitable for regression analysis.

Model Selection and Hyperparameter Tuning

A **Decision Tree Regressor** was chosen for this task, as decision trees are effective for handling both numerical and categorical data and are capable of capturing non-linear relationships. Hyperparameter tuning was performed using **GridSearchCV**, which exhaustively searched through various parameter combinations to find the best settings for the model. The parameters tuned included *max_depth*, *min_samples_split*, *min_samples_leaf*, and *criterion*.

Model Performance

The Decision Tree Regressor was trained using GridSearchCV, which performed exhaustive search across 360 different candidate hyperparameter combinations. After fitting 5 folds for each of these candidates, the optimal hyperparameters were identified as:

- **Criterion:** 'squared_error'
- **Max Depth:** 10
- **Min Samples Leaf:** 12
- **Min Samples Split:** 2

With these hyperparameters, the model achieved the following performance metrics:

- **Mean Squared Error (MSE):** 526,992.76
- **R-squared (R^2):** 0.879

The R-squared value of 0.88 indicates that the model is able to explain approximately 88% of the variance in sales data, showcasing a strong fit to the dataset. The Mean Squared Error indicates the average squared difference between the predicted and actual sales values, providing a quantitative measure of the model's prediction accuracy.

This strong performance suggests that the Decision Tree Regressor is an effective tool for predicting sales amounts based on the selected features, and can be useful for forecasting revenue and guiding business decisions.

Feature Importance

To better understand which features contributed the most to the predictions, feature importance was analyzed. The resulting plot revealed that certain features, such as PRICEEACH and QUANTITYORDERED, had a high impact on the predicted sales values, which makes sense as they directly influence the total sales amount.

Insights

The regression model provides a continuous prediction of sales, which can be particularly valuable for forecasting revenue, setting sales targets, and adjusting marketing strategies. By understanding the factors that most influence sales, businesses can make data-driven decisions to optimize their operations and sales strategies.